Chapter-2

• ENIAC (Electronic Numerical Integrator And Computer)

- ENIAC was the world's first general-purpose electronic digital computer.
- Built at the University of Pennsylvania and completed in 1946.
- Designed for the U.S. Army Ballistics Research Laboratory during World War II.
- Weighed 30 tons, occupied 1500 square feet, and used over 18,000 vacuum tubes.
- Consumed 140 kilowatts of power and performed 5000 additions per second.
- Operated using a decimal system; memory had 20 accumulators each holding a 10-digit decimal number.
- Programming was done manually using switches and cables.
- ENIAC marked the beginning of the electronic computer era.

• The Von Neumann Machine (IAS Computer) –

Stored-program concept: The idea that both data and instructions can be stored in main memory, making programming much easier and faster compared to manual wiring. Credited to John von Neumann and developed during the ENIAC era.

IAS Computer Structure: Became the model for later general-purpose computers. Main components:

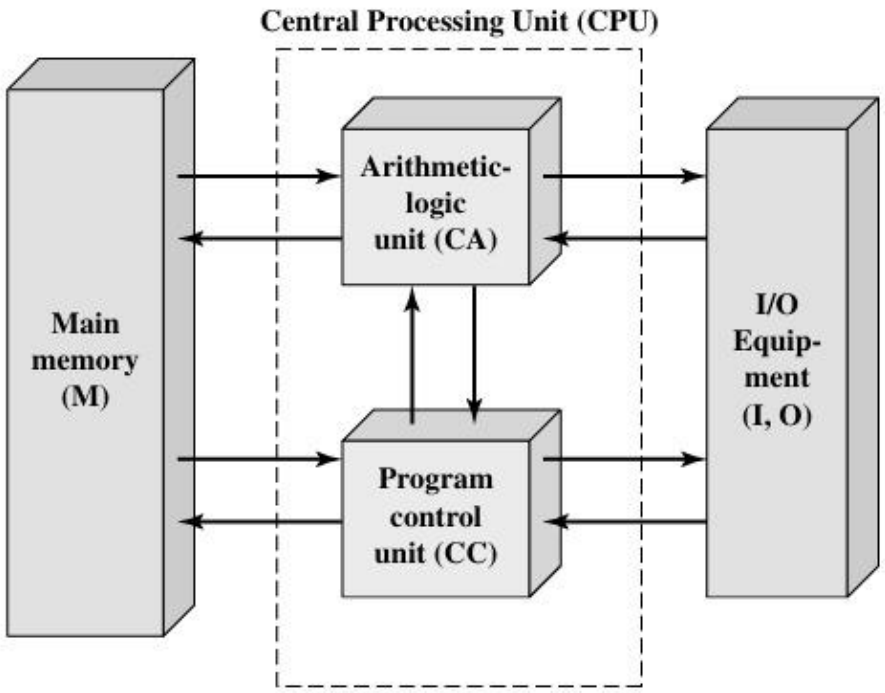

Figure 2.1 Structure of the IAS Computer

- **Main memory** (stores data and instructions)
- **Arithmetic Logic Unit** (ALU/CA: performs all arithmetic/logical operations)
- **Control Unit** (CC: interprets and sequences instructions)
- **Input/Output** (I/O: handles communication with external devices)

- **Process of the Von Neumann (IAS computer) architecture:**

- **Program and data stored in main memory:** Both the instructions (program) and data are kept together in the main memory.

- **Control unit reads instructions**: The control unit fetches instructions from memory, interprets them, and directs the necessary operations to the ALU or other units.

- **ALU performs operations**: The Arithmetic Logic Unit executes basic arithmetic (add, subtract, multiply, divide) and logic operations on the data as instructed.

- **Intermediate results and data stored**: Results of calculations, as well as data needed for further steps, are temporarily stored in memory.

- **Input/output operations**: Data and processed information move in and out of the system through I/O units, under the control unit's direction

- **Program sequence controlled**: The control unit ensures each instruction is processed in the correct order, managing the sequence and branching as needed (e.g., loops, conditions)

- **Memory Structure**

The IAS computer used a memory of 1000 storage locations called words.

Each word was 40 bits long (binary digits).

Both data and instructions were stored in these words in binary format.

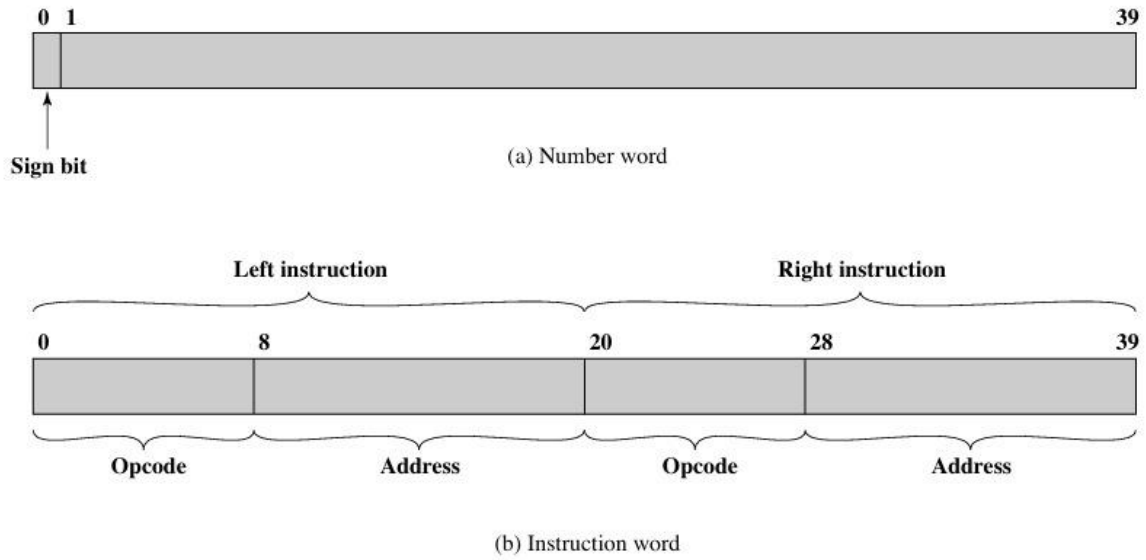

Figure 2.2 IAS Memory Formats

Number Format

A number word had a sign bit (1 bit) and a value field (39 bits).

Example: One word could represent a numeric value with a sign.

Instruction Format : Each word could also contain two instructions, "left" and "right," of 20 bits each

Each instruction consisted of:8-bit operation code (opcode) &12-bit address (pointing to a memory location).

Control Unit Operation

The control unit fetched instructions from memory and executed them one at a time

- **Types Of Register:**

Memory Buffer Register (MBR)

Holds a word to be stored in or received from memory, or sent to/received from the I/O unit.

Memory Address Register (MAR)

Specifies the address in memory for the word to be read or written, working with the MBR.

Instruction Register (IR)

Contains the current 8-bit opcode instruction being executed.

Instruction Buffer Register (IBR)

Temporarily holds the right-hand instruction from a memory word.

Program Counter (PC) Holds the address of the next instruction pair to be fetched from memory

Accumulator (AC) and Multiplier Quotient (MQ) Temporarily store operands and results of ALU operations. For multiplications, the AC stores the most significant bits and MQ stores the least significant bits.

IAS 21 instruction:

Table 2.1 The IAS Instruction Set

Instruction Type	Opcode	Symbolic Representation	Description
Data transfer	00001010	LOAD MQ	Transfer contents of register MQ to the accumulator AC
	00001001	LOAD MQ,M(X)	Transfer contents of memory location X to MQ
	00100001	STOR M(X)	Transfer contents of accumulator to memory location X
	00000001	LOAD M(X)	Transfer M(X) to the accumulator
	00000010	LOAD −M(X)	Transfer −M(X) to the accumulator
	00000011	LOAD |M(X)|	Transfer absolute value of M(X) to the accumulator
	00000100	LOAD −|M(X)|	Transfer −|M(X)| to the accumulator
Unconditional branch	00001101	JUMP M(X,0:19)	Take next instruction from left half of M(X)
	00001110	JUMP M(X,20:39)	Take next instruction from right half of M(X)
Conditional branch	00001111	JUMP+ M(X,0:19)	If number in the accumulator is nonnegative, take next instruction from left half of M(X)
	00010000	JUMP+ M(X,20:39)	If number in the accumulator is nonnegative, take next instruction from right half of M(X)
Arithmetic	00000101	ADD M(X)	Add M(X) to AC; put the result in AC
	00000111	ADD |M(X)|	Add |M(X)| to AC; put the result in AC
	00000110	SUB M(X)	Subtract M(X) from AC; put the result in AC
	00001000	SUB |M(X)|	Subtract |M(X)| from AC; put the remainder in AC
	00001011	MUL M(X)	Multiply M(X) by MQ; put most significant bits of result in AC, put least significant bits in MQ
	00001100	DIV M(X)	Divide AC by M(X); put the quotient in MQ and the remainder in AC
	00010100	LSH	Multiply accumulator by 2; i.e., shift left one bit position
	00010101	RSH	Divide accumulator by 2; i.e., shift right one position
Address modify	00010010	STOR M(X,8:19)	Replace left address field at M(X) by 12 rightmost bits of AC
	00010011	STOR M(X,28:39)	Replace right address field at M(X) by 12 rightmost bits of AC

• Second Generation: Transistors

- The transition from vacuum tubes to transistors made computers smaller, faster, cheaper, and more reliable.

- Transistors were fully adopted by the late 1950s, enabling mass production of systems and significant industry growth.

- This generation saw the introduction of new arithmetic and logic units, improved instruction sets, and higher-level programming languages.

- **IBM 700/7000 Series**
- IBM led the market with its 700 and 7000 series using both vacuum tube and transistor technology.

- Early models (like the 701) used vacuum tubes, while later ones (like the 7090 and 7094) used transistors with features such as core memory, index registers, and independent data channels for peripherals.

- Multiplexers and data channels allowed better CPU and I/O coordination and more efficient data handling

Third Generation: Integrated Circuits (IC)

- Integrated circuits (ICs) replaced discrete transistors, packing many components on a single silicon chip.

- ICs greatly reduced size, cost, and power consumption while increasing speed and reliability.

- Microelectronics led to computers with gates for logic (data processing) and memory cells for storage.

- This period (1960s onward) enabled mass-produced, more convenient, and portable computers with dramatic cost and performance benefits.

- **The characteristics of a computer family/360 Family**

Similar or identical instruction set: All family members support the same or nearly the same set of machine instructions, allowing easy migration of software within the family.

Similar or identical operating system: The same basic operating system is used for all members, though higher-end models may have extra features.

I**ncreasing speed:** Instruction execution rate increases from lower to higher family members.

Increasing number of I/O ports: More advanced models have more I/O ports.

Increasing memory size: Main memory size grows as you move to higher family members.

Increasing cost: More advanced systems cost more than simpler models at any given time.

- ## Intel processor evolution

8080: First general-purpose microprocessor, 8-bit, used in the Altair personal computer.

8086: More powerful, 16-bit microprocessor, wider data path, introduced instruction pipelining; 8088 variant used in IBM's first personal computer.

80286: Extended memory addressing up to 16 MB, compared to 1 MB of the 8086.

80386: First Intel 32-bit processor, allowed multitasking, improved speed and power.

80486: Added advanced cache technology, instruction pipelining, and built-in math coprocessor.

Pentium: Introduced superscalar technology, allowing multiple instructions to execute in parallel.

Pentium Pro: Improved on superscalar design, added register renaming, branch prediction, and data flow analysis.

Pentium II: Introduced MMX technology for better multimedia processing.

Pentium III: Added support for 3D graphics with new instructions.

Pentium 4: Added more floating-point instructions and multimedia features.

Core: First dual-core Intel x86 microprocessor, enabled two processors on a single chip.

Core 2: Extended architecture to 64 bits, supported four processors (Core 2 Quad) on one chip.

- **Embedded Systems**

- An embedded system combines hardware and software to perform a specific function, not general-purpose tasks.

- Examples: Car ignition systems, smart TVs, industrial robots, digital cameras, medical infusion pumps, and office printers

Features of Embedded Systems

- Used in everyday products and outnumber general-purpose computers.
- Requirements differ (size, cost, reliability, environment, performance needs)
- Often optimized for limited resources and long life with real-time processing

- **Structure of a EMB**

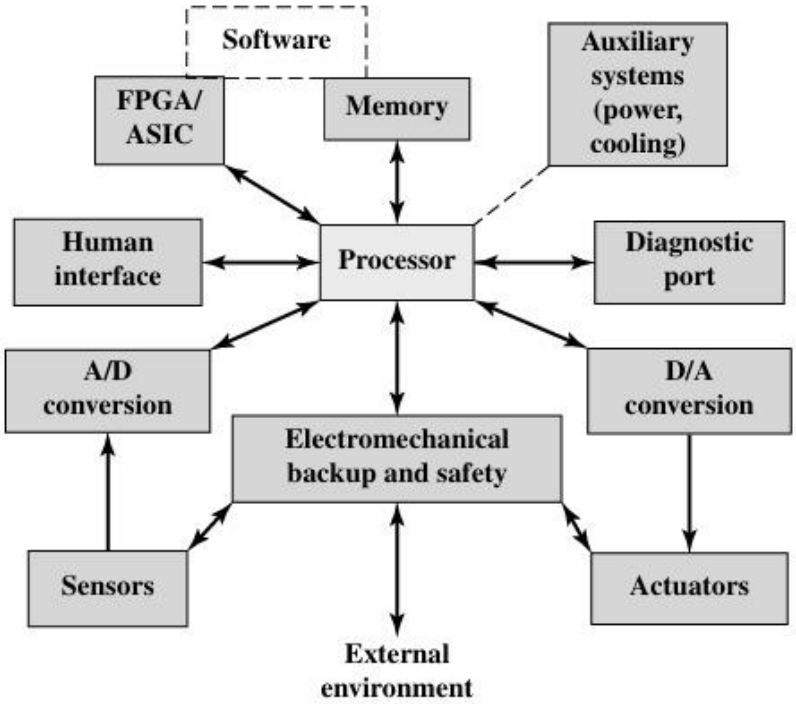

Figure 2.13 Possible Organization of an Embedded System

- **ARM Architecture**

ARM is a processor architecture based on RISC (Reduced Instruction Set Computing) principles.

Widely used in embedded systems due to its simple design, high efficiency, and low power consumption.

- **ARM Evolution**

ARM processors have evolved to offer better speed and more features with each generation.

Common in mobile, automotive, industrial, and consumer electronics for their balance of power, performance, and price.